

Integrated Control Policy for Heterogeneous Traffic in Container Terminals With Unsignalized Intersections

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Abstract—This research investigates the control difficulties related to heterogeneous traffic flow in container terminals, featuring both connected and automated vehicles (CAVs) and human-driven vehicles (HDVs). The lack of signal control at intersections and the unpredictable routes taken by HDVs make the efficient transport of containers within a terminal quite challenging. To tackle this issue, we present an integrated traffic control policy aimed at enhancing transportation efficiency. For each unsignalized intersection, a virtual token ring system is introduced to manage the passage of vehicles, using a back pressure-based algorithm and specific token delivery rules to determine phase sequence and duration. Furthermore, we introduce an improved back pressure-based dynamic routing method for CAVs, which allows for the selection of roads with shorter travel time while adhering to a travel distance constraint when crossing an intersection. This approach aims to minimize disruptions from HDVs, reduce travel time, and prevent excessively long travel distances. Multiple experiments are conducted to verify the proposed method's effectiveness.

Index Terms—Connected and automated vehicle, heterogeneous traffic flow, back pressure, unsignalized intersection, dynamic routing.

I. INTRODUCTION

CONNECTED and automated vehicles (CAVs) [1], [2] are progressively integrated into human environments, facilitating the efficient transportation of goods across an increasing number of industrial locations and thus driving improvements in transport infrastructure. Human-driven vehicles (HDVs) and CAVs collaborate in industrial transportation settings, forming heterogeneous transport systems [3]. For example, at container terminals, HDVs transport containers from the yard to widely dispersed customer locations, while

CAVs transport containers within the yard or between the yard and the quayside. However, high transportation activity within the yard often leads to container terminal congestion [4], reducing the efficiency of heterogeneous transport systems.

Some studies [5], [6] mitigate congestion by limiting the number of trucks arriving at the same time. Nonetheless, these methods are less ideal since they improve traffic conditions by decreasing terminal throughput. The study in [7] optimizes container distribution and balances the loading and unloading workloads of the terminal to prevent excessive truck clustering in specific areas. Despite their potential, these approaches encounter significant challenges when managing the increasing volume and diversity of containers. Furthermore, container terminal traffic networks continue to face challenges due to the erratic behavior of HDV drivers [8], who choose routes at intersections based on their random and unpredictable driving patterns. Additionally, unlike urban traffic grids, these networks generally lack traffic signals and primarily consist of single-lane roads, further increasing congestion.

To address these challenges, this study develops a container terminal congestion alleviation policy based on traffic management principles, integrating the Virtual Token Ring (VTR) intersection control method with a BP-based dynamic routing approach. The VTR is a virtual signaling system in which only the phase holding the token is granted the right-of-way at an intersection. In contrast to previous BP-based studies [9], [10], the proposed control policy considers both the head vehicle type and the queue length when determining the sequence and duration of token allocation. This approach improves intersection efficiency by mitigating conflicts between CAVs and HDVs, while also reducing vehicle waiting times. To further mitigate disruptions caused by HDVs and prevent potential local bottlenecks, the BP-based dynamic routing method incorporates the expected travel time as the BP weight for phases. Additionally, a heuristic function is introduced to minimize overall travel time and avoid excessively long travel routes.

The main contributions of this work can be summarized as follows:

- 1) We develop a VTR system to control the activation sequence and duration of each traffic phase at unsignalized intersections in the heterogeneous traffic network. Adhering to the token-delivery guidelines, vehicles can navigate intersections without conflicts.

Received 21 September 2024; revised 7 February 2025; accepted 12 March 2025. Date of publication 22 April 2025; date of current version 1 July 2025. This work was supported in part by the Science and Technology Program of Zhejiang Province of China under Grant 2023C01174, in part by the Foundation of Huaqiao University under Grant 605-50Y24033, and in part by Fujian Provincial Science and Technology Development Project Led by Central Government under Grant 2022L2012. The Associate Editor for this article was Y. Lv. (*Corresponding author: Weimin Wu.*)

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Digital Object Identifier 10.1109/TITS.2025.3560067

- 2) An innovative travel time-based dynamic routing method is proposed based on BP, dynamically updating CAV routes at each junction to mitigate disruptions from HDVs and minimize travel time.
- 3) We integrate the dynamic routing approach with a heuristic function to optimize both travel distance and time for CAVs.

The remainder of this paper is structured as follows: Section II reviews related work. Section III introduces the fundamental concepts and control problem for heterogeneous traffic flow. Section IV details the integrated control policy. Section V explains the experimental setup and analyzes the results. Lastly, Section VI concludes the paper and discusses future research directions.

II. RELATED WORK

Traffic management control methods designed to enhance throughput and alleviate congestion generally concentrate on managing intersections, optimizing vehicle routing, or integrating both methods. Regarding signalized intersections [11], [12], [13], current methods include fuzzy control [14], adaptive control [15], and neural network control [16], and others. These approaches analyze traffic flow patterns to predict future trends and dynamically adjust traffic signal timings or cycle lengths to enhance traffic conditions.

In the context of unsignalized intersections [17], [18], [19], some studies assume that all vehicles are either CAVs or equipped with vehicle-to-everything communication devices. Chen et al. [20] introduce a cooperative grouping method for CAVs to enhance the intersection throughput by mitigating conflicting movements. Xu et al. [21] project vehicles from multiple lanes into a virtual lane and design a conflict-free geometric topology to construct the virtual platoon. Additionally, a distributed controller is designed to stabilize the platoon, ensuring conflict-free cooperation at intersections. Hang et al. [17] utilize a Gaussian potential field for risk assessment, formulating a cost function based on vehicle risk and efficiency. This cost function is integrated into a game-theoretic decision-making framework to manage CAV coordination at unsignalized intersections. Nonetheless, the complexities arising from the interactions among various vehicle types complicate the control challenge. Consequently, numerous studies have neglected the diversity in traffic flow. For example, Li et al. [22] present a game-theoretic framework that simulates the interactive behavior of vehicles at uncontrolled intersections. Another study [23] introduces a unified safety-enhanced reinforcement learning and model predictive control framework to assist CAVs at unsignalized intersections. Yet, while these methods chiefly emphasize the advantages of CAVs, the safety and practicality of HDVs warrant further examination.

Dynamic routing is also employed to improve traffic conditions. The dynamic routing problem [24], [25], [26], [27] is also referred to as the traffic assignment problem. Soefker et al. [24] provide a comprehensive review of dynamic routing techniques aimed at mitigating congestion. Fu [28] proposes an adaptive routing method that utilizes real-time

information. Papadopoulos et al. [29] introduce a road pricing strategy that integrates pricing and routing schemes to reduce traffic congestion. In [30], a decentralized dynamic routing approach based on the back-pressure (BP) principle is introduced, offering guarantees of network-wide stability.

Moreover, various studies have merged traffic signal control with dynamic vehicle routing to better manage traffic networks. The study in [31] proposes a method to modify route selection based on travel time and origin-destination (OD) demands, while simultaneously optimizing signal timings and traffic routing. The approaches in [32], [33] aim to capture the interaction between vehicle routing and traffic signal control. Chai et al. [32] introduce an iterative method to enhance route plans using OD information, traffic signal schedules, and current travel times within the network. They then design an intersection signal control strategy to adjust the signal plan according to traffic flow distribution and queue length. Chen et al. [33] enable vehicles to collaborate in route decision-making to achieve user-optimal or system-optimal traffic states, while simultaneously modifying signal timing plans based on traffic volumes and vehicle waiting times. Regarding the connection of traffic networks, Souza et al. [34] extended the single-commodity traffic signal control strategy found in [35] into a multi-commodity traffic signal control and routing system. This novel method applies a store-and-forward model to manage traffic flow and formulates the traffic control challenge as a convex quadratic programming problem with linear constraints. Control decisions are rendered at each cycle using a rolling horizon control method to adapt to fluctuations. Drawing inspiration from multi-commodity BP algorithms, Zaidi et al. [36] introduced two BP-based algorithms aimed at managing signal phase activation and vehicle routing issues separately. These decisions are made locally with insights into adjacent queues, ensuring the algorithms remain decentralized and computationally efficient. A notable drawback of the BP-based approach is the potentially extensive delay when queues are short. As a response, Wu et al. [10] developed a method for selecting the active phase that accounts for fairness in delays and the Head-Of-Line vehicle delay data. Nevertheless, this algorithm is more appropriate for smaller scale networks.

In summary, the mentioned approaches are confined to traffic networks equipped with traffic signals or consisting solely of CAVs. The inherent unpredictability of HDVs disrupts traffic flow, making it challenging to accurately estimate traffic states and manage heterogeneous traffic flows. The BP algorithm is a decentralized traffic management method that utilizes real-time local traffic information, reducing computational complexity and eliminating the need for traffic pattern prediction. Accordingly, the following sections introduce a traffic planning method based on the BP algorithm.

III. PROBLEM DESCRIPTION

The traffic control issue at a container terminal is examined, featuring heterogeneous traffic flow. Furthermore, the intersections lack traffic lights, and the routes taken by HDVs are uncertain.

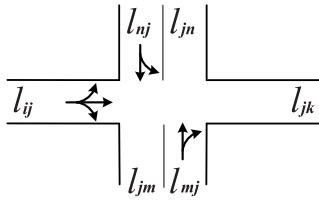


Fig. 1. An unsignalized intersection with six directed roads.

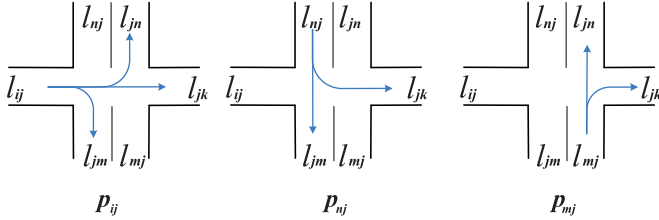


Fig. 2. Three phases of the intersection in Fig. 1.

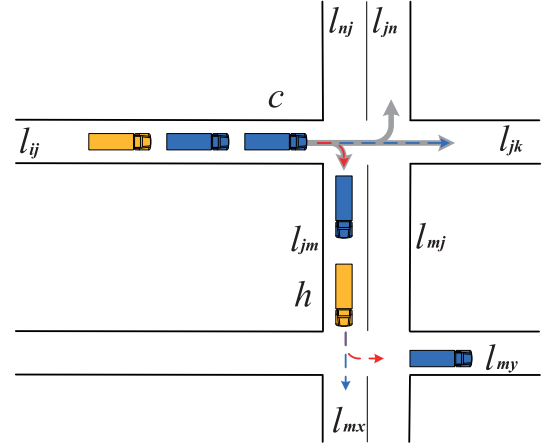
A. Transport Traffic Network

Given that a container terminal's traffic network typically resembles a grid [37], it can be modeled as a directed graph $G = (N, L)$, where N signifies the intersections and L represents the directed roads. The path between any two intersections i and j is symbolized by $l_{ij} \in L$, indicating the direction from intersection $i \in N$ to intersection $j \in N$ within the traffic network addressed in this study. The collection of vehicles within the network G is denoted by $C = C_h \cup C_c$, with C_h referring to the group of HDVs, C_c to CAVs, and $C_h \cap C_c = \emptyset$. A vehicle c travels from an origin road l_o^c to a destination road l_d^c ; let (l_o^c, l_d^c) represent the OD pair for a vehicle $c \in C$, where $l_o^c, l_d^c \in L$.

A traffic movement is defined as the passage of vehicles through an intersection, transitioning from an upstream road l_{ij} to a downstream road l_{jk} , and is represented by the pair (l_{ij}, l_{jk}) . A traffic phase, represented by p , comprises a collection of traffic movements that concurrently have the right of way at an intersection. Movements within the same phase are conflict-free, implying that no two movements in the phase intersect or interfere with each other. For instance, Fig. 1 illustrates an intersection j with six roads $l_{ij}, l_{jk}, l_{jn}, l_{nj}, l_{jm}$, and l_{mj} . Intersection j includes an upstream road set $\Omega_j = \{l_{ij}, l_{nj}, l_{mj}\}$, a downstream road set $R_j = \{l_{jk}, l_{jn}, l_{jm}\}$, three phases p_{ij}, p_{nj} , and p_{mj} . The three phases are depicted in Fig. 2, where $p_{ij} = \{(l_{ij}, l_{jk}), (l_{ij}, l_{jn}), (l_{ij}, l_{jm})\}$, $p_{nj} = \{(l_{nj}, l_{jm}), (l_{nj}, l_{jk})\}$, and $p_{mj} = \{(l_{mj}, l_{jk}), (l_{mj}, l_{jn})\}$. The state of a phase p_{ij} at time t is $p_{ij}(t) \in \{0, 1\}$. Specifically, $p_{ij}(t) = 1$ if the phase p_{ij} is active at time t , meaning the associated traffic movements of p_{ij} is active; otherwise, $p_{ij}(t) = 0$, indicating an inactive state.

B. Unsignalized Intersection

We assume that the traffic controller at the intersection operates in time $t \in \mathbb{Z}^+$, where $\mathbb{Z}^+$ is the set of non-negative rational numbers. In order to manage the varied traffic at each intersection, the following operations need to be carried out during each time slot t :

Fig. 3. An example of traffic congestion caused by HDV h . The red dotted line represents the original routing plan, and the blue dotted lines represent the alternative routing plan.

- The activation sequence is determined for each phase at the intersection;
- The duration of the active phase is determined based on the queue length of each phase at the intersection.

Furthermore, the lack of traffic signals at various intersections $j \in N$ will perplex human drivers navigating through them. The method for intersection control needs to consider both efficiency and the human element.

C. Disturbance From HDVs

The varying speed, acceleration, deceleration, and unpredictable route choices of HDVs [3], [38], [39] suggest that HDVs can be represented as uncertain vehicles, which can potentially cause moving bottlenecks. With these considerations, our research posits that HDVs typically move at slower speeds compared to CAVs. Moreover, given the disconnected and unpredictable behavior of HDVs, our control approach gives precedence to these vehicles at intersections. The objective is to aid human drivers in deciding when to cross intersections by applying simple rules while also boosting traffic flow efficiency through the management of CAVs. By giving priority to HDVs in this manner, we simplify their operation within the traffic network, thereby enhancing overall traffic management and safety.

A one-way road can be represented as a first-in-first-out system, where a preceding vehicle might obstruct the movement of those following it, leading to inefficiencies in the phase activation time. For instance, as demonstrated in Fig. 3, yellow vehicles denote HDVs while blue vehicles signify CAVs. The CAV c can choose among three traffic movements (l_{ij}, l_{jk}) , (l_{ij}, l_{jn}) and (l_{ij}, l_{jm}) , whereas the HDV h has two possible traffic movements (l_{jm}, l_{mx}) and (l_{jm}, l_{my}) available. We consider (l_{ij}, l_{jm}) as the initial next traffic movement for c , and (l_{jm}, l_{my}) as the subsequent traffic movement for h , with the respective phases p_{ij} and p_{jm} being activated. Nevertheless, road l_{my} is obstructed by other vehicles, and the driver of h sticks to the original route, causing insufficient storage capacity on the downstream road l_{jm} permitting further

movement. Consequently, the CAV c has to halt and wait, thereby impeding the vehicles in traffic movements (l_{ij}, l_{jk}) and (l_{ij}, l_{jn}) . In this scenario, if CAV c opts for (l_{ij}, l_{jk}) to replace (l_{ij}, l_{jn}) in its routing plan, both c and the other vehicles can proceed without delay.

Although each CAV in the heterogeneous traffic network has a fixed destination, due to the time-varying and stochastic nature of network flow, it is essential to design a flexible route for each CAV. This requires dynamically and adaptively selecting the optimal route for each CAV. Additionally, when planning the route, travel time and distance must be considered to ensure the chosen path is efficient, not just because it offers a shorter travel time.

D. Information of Vehicles

Road conditions are determined using inductive loops, video surveillance, and data collected from CAVs [40], [41], [42]. Additionally, queue lengths can be extracted from this information [43]. For the sake of simplicity, this research assumes the location and velocity of each vehicle are known. Moreover, virtual human drivers choose HDV paths arbitrarily.

IV. METHODOLOGY

The traffic control method based on the BP algorithm [44] serves as a decentralized system that effectively produces solutions grounded in instantaneous local traffic data. Importantly, this technique can be optimized while adhering to the network's overall capacity limits. Thus, the policy introduced in this study relies on the BP algorithm. Additionally, it comprises two main elements: the control method for unsignalized intersections and the dynamic routing method for CAVs. The specifics of the policy are outlined below.

A. BP-Based Unsignalized Intersection Control Method

The initial BP algorithm [36] typically activates the phase with the highest queue length during each time interval at signalized intersections. Nevertheless, at unsignalized intersections, the unpredictable phase sequence can lead to driver frustration and heightened collision risk. Additionally, when one inbound road encounters heavy traffic, others may be blocked for extended periods. To overcome these issues and efficiently regulate mixed traffic at unsignalized intersections, we propose a VTR system rooted in the BP algorithm to manage traffic flow at these junctions.

Originally, the token ring protocol [45] was created to eliminate access conflicts in local area networks by allowing only nodes with the token to send data, avoiding possible collisions. We have now applied the token ring concept to manage unsignalized intersections. In each intersection, a token is used to regulate the order of traffic phases, thus avoiding conflicts among various traffic streams.

An independent VTR system is made up of several stations and a single token. Fig. 4 illustrates an independent VTR system at intersection j , where $j \in N$. This system includes a series of stations and a token, known as the station set $S_j = \{s_{ij}, s_{nj}, s_{mj}\}$. Each station s corresponds to a unique traffic phase p and is depicted as a hexagon. Additionally,

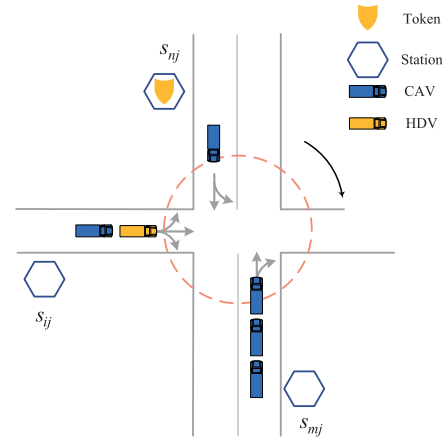


Fig. 4. A VTR system of the intersection j . Station s_{nj} holds the token.

stations are sequentially arranged in a clockwise order in list $S_j = \{s_{ij}, \dots, s_{|S|}, s_{ij}, \dots, s_{|S|}\}$ to control the sequence of traffic phases. The single token, shown as a yellow shield, is crucial in the VTR system, as only the station holding the token can activate its respective phase. For example, as shown in Fig. 4, station s_{nj} has acquired the token, thereby activating the corresponding traffic phase p_{nj} , where $p_{nj} = s_{nj} = 1$. Activating this phase allows the traffic flow $\{(l_{nj}, l_{jm}), (l_{nj}, l_{jk})\}$ to move through the intersection.

It should be noted that directing traffic at intersections without signals involves distinct difficulties due to the lack of signal devices and the involvement of HDVs. To tackle these issues and improve the effectiveness of the mixed traffic environment, both HDV operators and CAVs need to follow these token delivery guidelines:

- 1) The token is passed in a clockwise order.
- 2) The phases where the first vehicle in the queue on the road l_{ij} is an HDV are prioritized.

Denote $\gamma_{ij} \in 0, 1$ as the queue classification of the phase. When the phase pertains to a lane led by an HDV, it has priority, denoted by $\gamma_{ij} = 1$. If not, then $\gamma_{ij} = 0$.

Previous research [10], [30] indicates that the green time for traffic phases is traditionally set to a fixed duration. This approach might not be effective for managing mixed traffic, especially when there is an imbalance in the number of vehicles across different roads. Hence, this section proposes an adaptive phase duration by featuring a dynamic cycle length. The length of each phase is calculated using the BP algorithm. The BP value is derived from the smaller number between the queue length of the upstream road and the remaining capacity on the downstream roads. Concretely, let $q_{ij}(t)$ represent the queue length on the upstream road l_{ij} at time slot t , and let $z_{jk}(t)$ indicate the remaining capacity on the downstream road l_{jk} during the same time slot. For traffic flow from road l_{ij} to road l_{jk} during phase p_{ij} , the BP value is defined as follows:

$$BP_{ij,jk}(t) = \min\{q_{ij}(t), z_{jk}(t)\} \quad (1)$$

Taking into account the difference in length of each road, the average BP for movement (l_{ij}, l_{jk}) is considered and

calculated as follows:

$$BP_{ij,jk}^*(t) = \frac{\min\{q_{ij}(t), z_{jk}(t)\}}{|l_{ij}|} \quad (2)$$

where $|l_{ij}|$ is the length of road l_{ij} . Hereafter, the BP of the phase p_{ij} is:

$$BP_{ij}^*(t) = \frac{\min\{q_{ij}(t), \sum_{jk \in R_{ij}} z_{jk}(t)\}}{|l_{ij}|} \quad (3)$$

where R_{ij} is the downstream road set of l_{ij} . Then, the weight for phase p_{ij} is given by:

$$w_{ij}(t) = \max\{BP_{ij}^*(t), 0\} \quad (4)$$

The token processing time for station s_{ij} , which corresponds to the duration of phase p_{ij} , is denoted as t_{ij}^* . If $w_{ij}(t)$ is greater than 0, it means that vehicles are queuing on l_{ij} . Conversely, if there is no vehicle queuing, $w_{ij}(t) = 0$, and subsequently $t_{ij}^* = 0$. The BP weight is utilized to determine the token processing time. In detail, the cycle length is denoted as T , the processing time is calculated as follows:

$$t_{ij}^* = \begin{cases} \frac{w_{ij}(t)T}{\sum_{ij \in \Omega_j} w_{ij}(t)} & \text{if } w_{ij}(t) > 0 \\ 0 & \text{if } w_{ij}(t) = 0 \end{cases} \quad (5)$$

Furthermore, station s_{ij} will immediately release the token once the token holding time is exhausted and the corresponding phase p_{ij} is deactivated. Then, the token circulates in a clockwise direction to find the next appropriate station, ensuring that the phases at an intersection are activated in sequence.

Note that $s_{ij}(t) = 1$ and $p_{ij}(t) = 1$ when the phase is activated at time t . Since phases cannot be activated simultaneously, the following constraint must be satisfied:

$$\sum_{p_{ij} \in P_j} p_{ij}(t) = 1 \quad (6)$$

$$\sum_{s_{ij} \in S_j} s_{ij}(t) = 1 \quad (7)$$

Additionally, since external trucks are human-driven, the phase switching time must account for the continuous passage of these trucks in a queue. If the first vehicle in the queue on the upstream road corresponding to the phase is still an external truck after the initial activation time, the current phase should be maintained to allow their passage. The method for adjusting the phase activation time is detailed in Algorithm 1. The algorithm has three main inputs: the initial activation time of a specific traffic phase p_{ij} , denoted as t_{ij}^* ; the queue of trucks on the corresponding lane l_{ij} ; and the unit time increment $\bar{\tau}$. The output of the algorithm is the adjusted activation time of the traffic phase p_{ij} , denoted as t_{ij}^* .

The execution of Algorithm 1 begins with a loop that dynamically adjusts the activation time of the traffic phase by assessing the traffic conditions on the road l_{ij} . After the current activation time t_{ij}^* elapses, the algorithm checks if the leading truck in the queue on lane l_{ij} is an HDV. If so, the algorithm extends the activation time by adding a fixed unit

of time $\bar{\tau}$ to accommodate the passage of HDVs. After iterative evaluations and adjustments, the algorithm finalizes and returns an optimized phase activation time t_{ij}^* . This adjustment mechanism makes Algorithm 1 flexible for managing traffic flows with a higher proportion of manually driven trucks, improving traffic flow efficiency and significantly reducing congestion and delays.

Algorithm 1 Dynamic Cycle Period Phase Activation Time Algorithm

Input : Initial activation time t_{ij}^* for phase p_{ij} , the queue of trucks on lane l_{ij} , and the unit time increment $\bar{\tau}$.

Output: Adjusted activation time t_{ij}^* for phase p_{ij} .

```

1 while True do
2   if the first truck in the queue on lane  $l_{ij}$  is a CAV
   after time  $t_{ij}^*$  then
3     Keep activation time unchanged; break.
4   if all trucks on lane  $l_{ij}$  have passed within time  $t_{ij}^*$ 
   then
5     Keep activation time unchanged; break.
6   else
7     Increase activation time by one time unit:
        $t_{ij}^* = t_{ij}^* + \bar{\tau}$ .
8 return  $t_{ij}^*$ 

```

According to the above notions and preparations, the intersection control method that integrates VTR and BP is outlined in Algorithm 2. The input of the algorithm includes S_j and the last token-holding station $\tilde{s}$ from the previous cycle. The output specifies the token delivery sequence and the corresponding token duration for the current cycle. Steps 1 - 2 arrange the stations in list S_w in a clockwise direction based on the station $\tilde{s}$. Initialize an empty list $S_j^*(t)$ to store the sorted stations for the subsequent stages. Step 3 calculates all phase weights of the intersection j . Steps 4 - 6 identify the queues in S_w for which the leading vehicle is an HDV, and append these stations to the list $S_j^*(t)$. Steps 7 - 9 evaluate the weight values of the remaining stations, and if its value is not 0, append them to the list $S_j^*(t)$ in descending order. Steps 10 - 12 calculate the token-holding time of every station in the $S_j^*(t)$ by Eq. (5) and append the result to list $T_j^*(t)$ in order.

Example 1: As shown in Fig. 4, three stations s_{ij} , s_{nj} , and s_{mj} are depicted, all originating from intersection j . Assume that in the previous cycle, the last holding token station was $\tilde{s} = s_{ij}$. According to Steps 1 - 2 of Algorithm 2, we determine $S_w = [s_{nj}, s_{mj}, s_{ij}]$ and $S_j^*(t) = \emptyset$. Subsequently, through Step 3, we can get the weight value set $W_j(t) = \{\frac{1}{3}, \frac{1}{2}, \frac{1}{6}\}$. Since the first vehicle in the queue of road l_{ij} is a HDV, i.e., $\gamma_{ij} = 1$, we put s_{ij} into set $S_j^*(t)$ according to Steps 4 - 6. Following Steps 7 - 9, both s_{mj} and s_{nj} are added to list $S_j^*(t)$. With an assumed intersection cycle time of 30 seconds, Steps 10 - 12 provide the duration list $T_j^*(t) = \{10, 15, 5\}$.

Algorithm 2 BP-Based Unsignalized Intersection Control Algorithm

Input : The list S_j with $2 \times |S|$ stations; the last holding token station $\tilde{s}$;

Output: Activate sequence set $S_j^*(t)$; activation time set T_j^* ;

- 1 Find the first element $\tilde{s}$ in the list S_j , and place its subsequent $|S|$ elements in list S_w ;
 - 2 Set the list $S_j^*(t) = \emptyset$;
 - 3 For any $s_{ij} \in S_w$, calculate the weight w_{ij} of their corresponding phase p_{ij} by Eqs. (3) and (4), and put all weights in the set $W_j(t)$;
 - 4 **foreach** $s_{ij} \in S_w$ **do**
 - 5 **if** $\gamma_{ij} == 1$ **then**
 - 6 append $s_{ij}(t)$ to $S_j^*(t)$;
 - 7 **while** $S_w \neq \emptyset$ **do**
 - 8 **if** $w_{ij} \neq 0$ **then**
 - 9 append $s_{ij}(t)$ into $S_j^*(t)$;
 - 10 **foreach** $s \in S_j^*(t)$ **do**
 - 11 Determine the duration time t_{ij} by $W_j(t)$ and Eq. (5);
 - 12 Put $t_{ij}(t)$ into $T_j^*(t)$;
 - 13 **return** $S_j^*(t), T_j^*(t)$
-

B. BP-Based Dynamic Routing Method

As mentioned in Section II, the uncertainty caused by HDVs in heterogeneous traffic networks leads to fluctuations in traffic flow, thereby affecting the transport efficiency of CAVs. In this section, we design a BP-based dynamic routing method for CAVs, enabling them to adjust their route plans at each intersection according to the BP weight of subsequent downstream roads to mitigate potential congestion. Due to the varying spatial distribution of vehicles along roads, the BP approach based on queue length may not accurately reflect road traffic capacity [46]. Therefore, we employ the estimated travel time as pressure weight in the dynamic routing method. In this case, considering the lower average speed of HDVs, the travel states can be classified into two categories: the free-flow state and the constrained state. In the constrained state, the travel time is limited by the speed of the preceding vehicle. Let $t_{jk}^f(c)$ denote the estimated flow time of a CAV $c \in C_a$ on the road l_{jk} . It is given by:

$$t_{jk}^f(c) = \max_{c' \in C_{jk}} \left\{ \frac{|l_{jk}| - |l_{jk}'(t)|}{v_{c'}(t)} + \frac{|l_{jk}'(t)|}{v_c(t)} \right\} \quad (8)$$

where C_{jk} is the set of vehicles on road l_{jk} , $v_c(t)$ is the speed of vehicle c , $v_{c'}(t)$ is the speed of vehicle c' , $|l_{jk}|$ is the length of road l_{jk} , and $|l_{jk}'(t)|$ represents the remain travel distance of vehicle c' on road l_{jk} at time t .

The BP of road l_{jk} for c is computed as follows:

$$BP_{jk}^c = \frac{|l_{jk}|}{t_{jk}^f(c)}, \quad l_{jk} \in R_{ij} \quad (9)$$

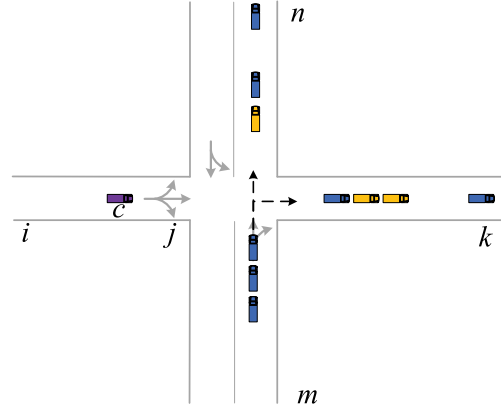


Fig. 5. Possible intersections for vehicles on the road l_{ij} .

Then, the pressure weight is as follows:

$$w_{jk}^c = BP_{jk}^c, \quad l_{jk} \in R_{ij} \quad (10)$$

The vehicle c at intersection j chooses the subsequent route r_j^c to optimally alleviate pressure, i.e.,

$$r_j^c = \arg \max_{l_{jk} \in R_{ij}} \{w_{jk}^c\} \quad (11)$$

Therefore, CAVs can be guided to a faster road by the proposed dynamic routing approach, which significantly reduces the possibility of vehicle accumulation and minimizes potential congestion. Nevertheless, the proposed approach only explores the immediate future well, which may result in CAVs traveling longer distances. However, this may not be optimal for the overall transportation system. To address this issue, we propose a travel distance and time balance method inspired by the A* algorithm [47], which is used to search the shortest route between the origin and destination effectively. A typical A* algorithm determines the next road by selecting the road that minimizes the cost, with the cost function defined as follows:

$$f_c(k) = g_c(k) + h_c(k), \quad k \in N \quad (12)$$

where $f_c(k)$ represents the estimated distance from the origin to the destination of vehicle c , $g_c(k)$ is the distance from the origin to the intersection k , i.e., the cumulative travel distance, and $h_c(k)$ is a heuristic function that estimates the possible shortest path from intersection k to the destination. If $h_c(k)$ is admissible, the estimated distance is never longer than the distance to reach the destination.

Considering the presence of diagonal paths in some container terminals, this study employs Euclidean distance as the heuristic distance $h_c(k)$. As depicted in Fig. 5, if a CAV c is on the road l_{ij} , it must navigate to its destination through one of the following downstream intersections: n , k , or m . We let $f_c(k)$ denote the estimated distance of c via downstream intersection k from origin to destination. Using the cost function, we can calculate the shortest route via the intersection j of a CAV c :

$$f_c'(j) = \min\{f_c(k)\}, \quad l_{jk} \in R_{ij} \quad (13)$$

For the purpose of having more routing options and shorter routes, we add a parameter relaxation η_c to relax the travel distance constraint, as shown in the following expression:

$$f_c^*(j) = f_c'(j) + \eta_c \quad (14)$$

where $f_c^*(j)$ represents the relaxed cost function for CAV c at intersection j , and η_c is a non-negative real number. A binary variable is adopted to determine whether the next routing road satisfies the new distance constraint:

$$\lambda_{jk}(c) = \begin{cases} 0, & f_c(k) > f_c^*(j) \\ 1, & f_c(k) \leq f_c^*(j) \end{cases} \quad (15)$$

where $l_{jk} \in R_{ij}$. As shown in Eq. (15), the constraint associated with the road l_{jk} is denoted by $\lambda_{jk}(c)$. Initially, when c is at intersection j , it selects the next road r_j^c based on the highest pressure release. This selection can now be reformulated as follows:

$$r_j^c = \arg \max_{l_{jk} \in R_{ij}} \{w_{jk}^c \lambda_{jk}(c)\} \quad (16)$$

Therefore, the new next route r_j^c is determined, and the corresponding next road is l_{js} . Then, η_c is updated as follows:

$$\eta_c = \max\{\eta_c - (f_c(s) - f_c'(j)), 0\} \quad (17)$$

where $f_c(s)$ is the cost function of the next selected road.

Algorithm 3 BP-Based Dynamic Routing Method for CAV

Input : The downstream roads set R_{ij} ; distance relaxation parameter η_c ; the downstream road pressure weight set W_{ij}

Output: Next routing road l_{js} ; updated η_c

```

1 foreach  $l_{jk} \in R_{ij}$  do
2    $\lfloor$  Calculate and put  $f_c(k)$  into set  $F_{ij}$ 
3   Get  $f_c'(j)$  by Eq. (13);
4   Get  $f_c^*(j)$  by Eq. (14);
5   foreach  $f_c(k) \in F_{ij}$  do
6     if  $f_c(k) > f_c^*(j)$  then
7        $\lfloor \lambda_{jk}(c) = 0;$ 
8     else
9        $\lfloor \lambda_{jk}(c) = 1;$ 
10     $\lfloor$  put  $\lambda_{jk}(c)$  into set  $\Lambda_j$ ;
11 foreach  $l_{jk} \in R_{ij}$  do
12    $\lfloor$  Update  $w_{jk}^c \in W_{ij}$  by Eq. (16) based on  $\Lambda_j$ ;
13 Let the max  $w_{js}^c$  of set  $W_{ij}$  corresponding route as new  $r_j^c$ ;
14 Get road  $l_{js}$  according to  $r_j^c$  as the next routing road;
15 Update  $\eta_c$  by Eq. (17);
16 return  $l_{js}, \eta_c$ 

```

Algorithm 3 summarizes the procedure to update the routing plan for the CAV. In Algorithm 3, we apply the A* algorithm along with the parameter η_c to modulate the strictness of this shortest path constraint. Alternative routes within the η_c

limit can be selected based on congestion states. This adjustment allows for controlled vehicle travel distances, avoiding reliance on the shortest path in congested traffic scenarios. Consequently, our BP-based dynamic routing method directs vehicles along shorter paths and considers faster, less congested alternatives, embodying the ability to adapt to evolving traffic conditions.

V. EXPERIMENTS

To evaluate the effectiveness of the proposed BP-based intersection control and dynamic routing policy (IR-BP), we assess its performance by comparing it with other traffic control approaches using the microscopic traffic simulation tool Simulation of Urban Mobility (SUMO) [48].

A. Network and Demand

This section examines the heterogeneous traffic control problem in a container terminal transportation system. We use a traffic network model based on a simplified real-world container terminal as our test network. Distinct from typical urban traffic scenarios, most roads in the grid traffic network of the container terminal are single-lane roads. Specifically, the employed longitudinal roads are two-way, and most of the lateral streets are one-way. The layout of the container terminal is depicted in Fig. 6, consisting of a four-by-five grid with 20 unsignalized intersections and 54 roads of varying lengths, ranging from 45 to 300 meters. HDVs enter and exit the network through two gates located at the network's periphery, while CAVs are confined to roads within the container terminal. For HDVs, the OD demands involve traveling from a gate to an internal road, or conversely, whereas the OD demands for CAVs involve traveling between different terminal roads. These OD demands are generated randomly to emulate varied traffic conditions, and three different demand levels are assessed: 1600 demand/h (level 1), 2000 demand/h (level 2), and 2400 demand/h (level 3). These travel demands are progressively loaded into the simulations at five-minute intervals. According to their specific OD demands, HDVs follow the shortest route policy, selecting their routes at every intersection. Additionally, they randomly choose alternative routes with a probability of 20% to emulate the arbitrary driving behavior of human drivers. The maximum speed for HDVs varies between 9 and 12 m/s, while the maximum speed of CAVs is 14 m/s.

B. Compared Methods

Due to limited research on specific control methods for unsignalized intersections in heterogeneous traffic environments, our simulations assume that traffic lights are employed at the intersections when evaluating the comparison methods. We compare the performance of the proposed IR-BP policy with four different methods:

- 1) Fixed signal control with shortest travel time routing (FX-STR): The traffic signals at each intersection are activated on a predefined cycle. CAVs dynamically select their shortest travel time routes to reach their respective destinations.

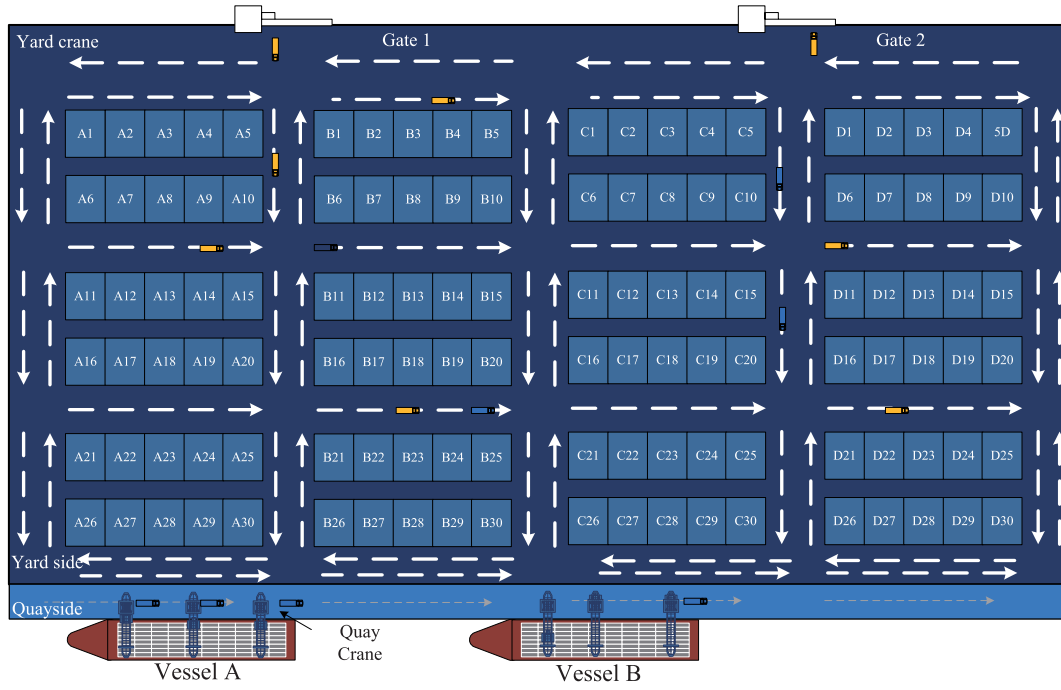
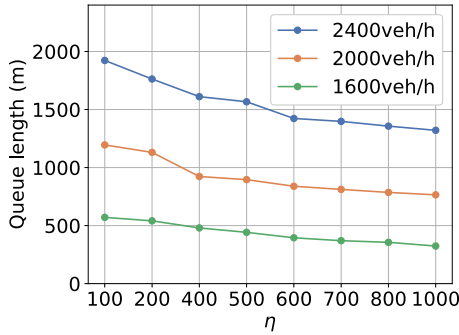
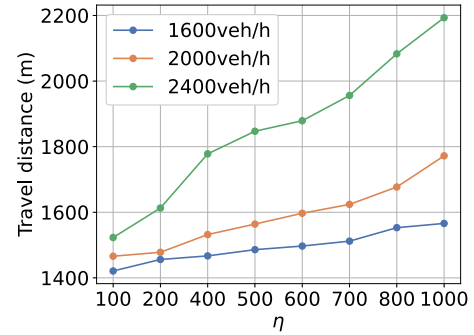


Fig. 6. Layout of the container terminal network.

Fig. 7. Vehicle queueing for different η .Fig. 8. Travel distance for different η .

- 2) Multi-commodity back-pressure control (MC-BP) [36]: A multi-commodity BP scheme for traffic signal control, with fixed routing for vehicles. In this method, CAVs follow the shortest routes to their respective destinations.
- 3) Adaptive routing back-pressure control (AR-BP) [36]: A multi-commodity BP algorithm for traffic signal control, while CAVs follow a dynamic routing strategy.
- 4) Multi-commodity formulation with integrated signal control and routing (MCSR) [34]: A real-time policy that integrates traffic signal control with multi-commodity routing for CAVs in networks with multiple destinations. The policy optimizes signal timings during every cycle and produces vehicle-specific routing decisions.

In the comparison method, each signal phase has an equal duration of 20 seconds for green light. Container terminal congestion is a primary cause of increased transportation time. Therefore, traffic congestion-related performance indicators are used to evaluate the effectiveness of the proposed method.

C. Simulation Results

In this section, we first analyze the influence of the distance relaxation parameter η on the performance of the proposed method. Simulations are conducted under diverse traffic demand scenarios using the IR-BP control method with various η values. The simulation results are plotted in Figs. 7 – 9. Fig. 7 shows that as η increases, the network queue length consistently decreases in all scenarios. This indicates that a larger relaxation parameter η offers CAVs a higher likelihood of selecting alternative routes to avoid congested roads, thus reducing network congestion. Fig. 8 illustrates that the travel distance of vehicles increases as η increases, suggesting that the route plan can be flexibly adjusted by η . Different from Figs. 7 and 8, Fig. 9 exhibits a non-monotonic trend in travel time. As η increases, the travel time initially decreases; however, it starts to rise as η approaches 500, potentially due to the rerouting of CAVs to longer paths, thereby leading to increased travel times. In other words, the additional relaxation variable can be employed to

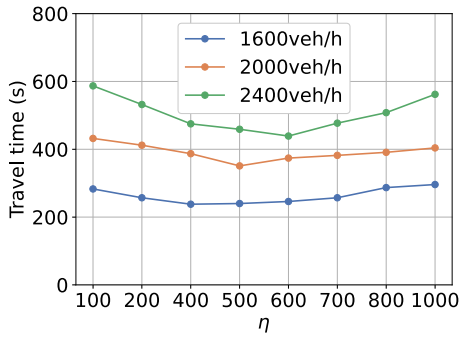


Fig. 9. Travel time for different η .

balance the travel distance and time in a heterogeneous traffic environment. However, an excessively large value of η will diminish the effectiveness of these adjustments on travel time. Therefore, we set $\eta = 500$ in the following experiments.

In this part, we evaluate the performance of the proposed IR-BP policy against four other methods across three distinct traffic demand scenarios, with an HDV penetration rate of 10%. Traffic network congestion performance is assessed using vehicle queue length, defined as the cumulative length of all vehicle queues throughout the network. The simulation results under level 1 traffic demand are given in Fig. 10(a). The queue lengths remain low for all control methods, indicating the comparable performance of all tested control methods under low-level traffic demand. However, as shown in Fig. 10(b), increased traffic demand results in a corresponding rise in queue lengths for the FX-STR and MC-BP methods. This indicates that the network performance drops significantly when the traffic demand exceeds load capacity in these two methods. In contrast, the AR-BP, DSP, and IR-BP methods show less performance degradation. Notably, both DSP and IR-BP achieve significantly lower congestion lengths compared to alternative methods. Among these two approaches, IR-BP demonstrates superior performance, effectively reducing queue lengths in the network and mitigating the spread of congestion. Moreover, the experimental results indicate that fixed signal control and shortest-route methods tend to intensify traffic congestion.

Furthermore, the average travel distance, travel time, and speed of vehicles under various demand scenarios are evaluated and depicted in Fig. 11. Here, travel distance and time are defined as the total distance and time required to complete a transportation task. As shown in Fig. 11(a), the MC-BP method achieves the shortest travel distance compared to dynamic routing methods, as it adheres to the initially planned shortest route. However, Figs. 11(b) and (c) indicate that MC-BP results in longer travel times and lower travel speeds than other methods. This suggests that although the MC-BP method has the shortest travel distance, it results in a longer travel time and a lower average speed. This could be attributed to fixed-route methods causing vehicle accumulation, leading to congestion and longer waiting times on the road. In contrast, the DSP method utilizes a multi-commodity model to optimize signal phase transitions and vehicle routing, thereby reducing driving distance and shortening travel time,

which demonstrates that flexible routing methods help alleviate congestion. As shown in Fig. 11(c), the travel speed of IR-BP and other adaptive dynamic routing methods is higher than that of MC-BP. This advantage is attributed to the ability of these dynamic routing methods to avoid heavily congested roads. Although the DSP method achieves shorter travel distances, its effectiveness in reducing travel time is relatively limited. This limitation may arise from the limited adaptability of the route-planning strategy to diverse routing demands and its inefficiency in utilizing road resources, which still results in prolonged vehicle waiting times. Fig. 11(b) displays the increase in travel time with increasing travel demand for all methods. However, the performance of IR-BP is better than that of other methods, especially at higher demand levels. Notably, while CAVs under the IR-BP control policy may follow longer routes, their overall travel times are significantly lower, and their speeds are higher compared to other methods. This reduction in average travel time is attributed to the shorter queue length in the network, as observed in Figs. 10(b) and (c).

Moreover, to evaluate the efficacy of the proposed method under different HDV penetration rates, the simulation scenario is based on level 2 traffic demand with varying HDV penetration levels. Specifically, HDV penetration rates are set at 20%, 30%, and 50%. The corresponding results are presented in Fig. 12. As expected, the results indicate that queue length increases with higher HDV penetration rates across the three comparison scenarios. A rising trend in queue length for the FX-STR, MC-BP, and AR-BP methods is evident in both Fig. 10(b) and Fig. 12. Notably, at lower HDV penetration rates, the queue length of the DSP method is shorter than that of the AR-BP method. However, as the penetration rate increases from 20% to 50%, the queue length under the DSP method surpasses that of AR-BP. Although the DSP method employs global optimization, its effectiveness in reducing congestion is constrained by the FIFO constraints and the unpredictable behavior of HDVs. In contrast, the IR-BP method effectively reduces queue accumulation in these scenarios.

Additional metrics are compared in Fig. 13. With the increase in HDV penetration, both average travel distance and time increase. Since HDVs cannot dynamically adjust their routes in response to real-time traffic conditions, CAVs are more likely to be obstructed by HDVs. This trend worsens as traffic volume increases, leading to a reduction in vehicle speeds and system throughput. As shown in Fig. 12(c), the rise in HDV numbers results in longer queues, diminishing the effectiveness of congestion mitigation methods. The control method proposed in this study shows better performance than other methods. Since HDVs and CAVs follow uniform token delivery rules, they can traverse intersections without conflict and with higher efficiency. The utilization of the BP-based token holding time control method further reduces wasted time. Concurrently, implementing a travel time BP-based dynamic routing method allows for timely modification of subsequent routes based on expected travel time, substantially mitigating disruptions from HDVs. However, an increase in the number of HDVs will inevitably lead to a decline in the efficiency of the traffic system.

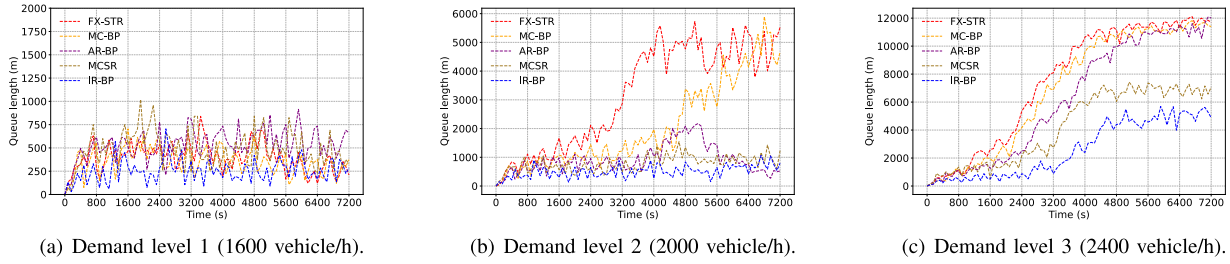


Fig. 10. Vehicle queueing for different demand levels.

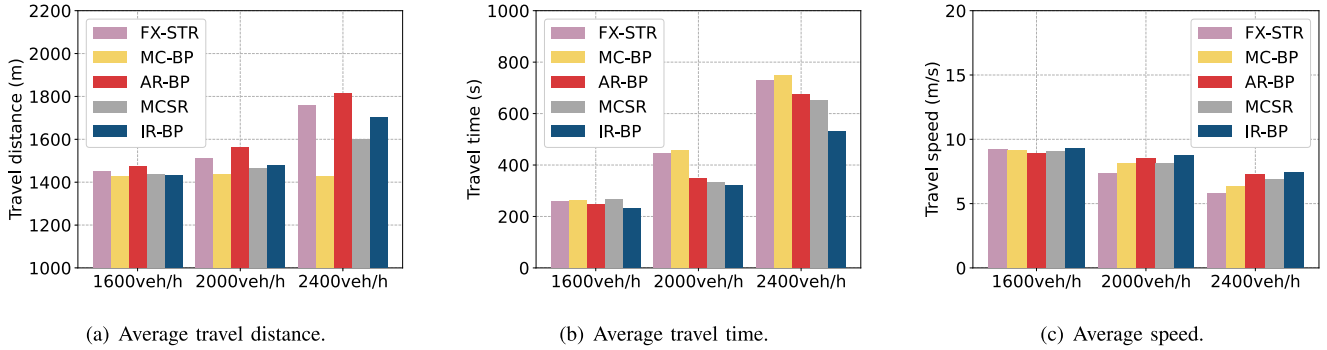


Fig. 11. Simulation results for different demand levels.

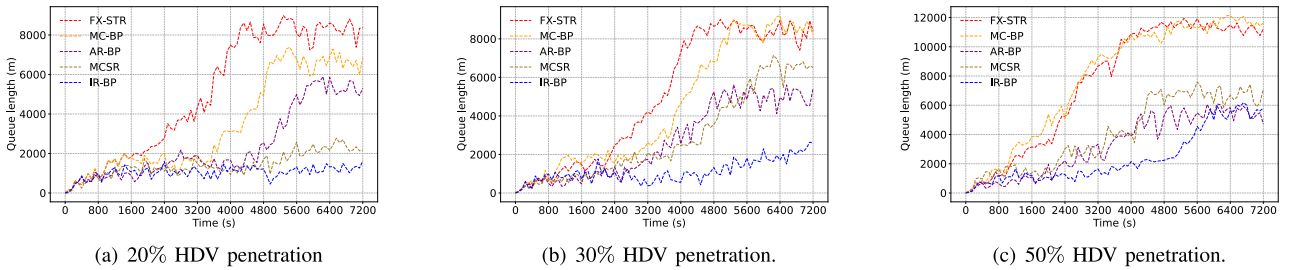


Fig. 12. Vehicle queueing for different HDV penetrations.

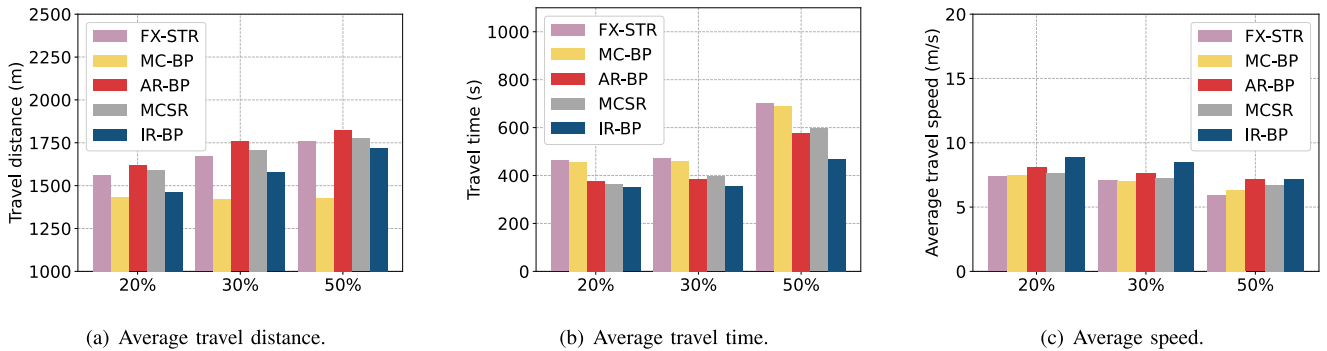


Fig. 13. Simulation results for different HDV penetrations.

The spatial and temporal distribution of transportation demand significantly impacts vehicle transportation efficiency. To verify the stability of the proposed method, we conduct experiments on eight different cases that are extracted and simplified from an actual container terminal. These cases involve transportation tasks for CAVs between container yards and quaysides and transportation tasks for HDVs between gates and yards. These tasks focus on the origin and destination points of the transportation tasks, and their respective start

times. Table I presents data on the average travel distance, travel time, and waiting time for CAVs under the control of MCSR and IR-BP. Here, waiting time refers to the total duration spent waiting during a transportation task. Furthermore, Fig. 14 presents the corresponding queue lengths for each case.

The performance of the two methods varies across different transportation cases. Overall, the IR-BP control method shows better performance than the MCSR method. Table I indicates that the IR-BP control method achieves shorter average travel

TABLE I
TRAVEL TIME, TRAVEL DISTANCE, AND WAITING TIME FOR EIGHT CASES UNDER THE CONTROL OF MCSR AND IR-BP

Case	MCSR			IR-BP		
	travel time(s)	travel distance(m)	waiting time(s)	travel time(s)	travel distance(m)	waiting time(s)
1	407	1373	139	335	1327	92
2	373	1360	121	327	1333	84
3	401	1454	164	371	1412	121
4	468	1432	132	383	1384	126
5	385	1361	131	333	1312	78
6	438	1429	165	337	1341	114
7	392	1358	140	317	1370	98
8	415	1415	174	325	1340	106

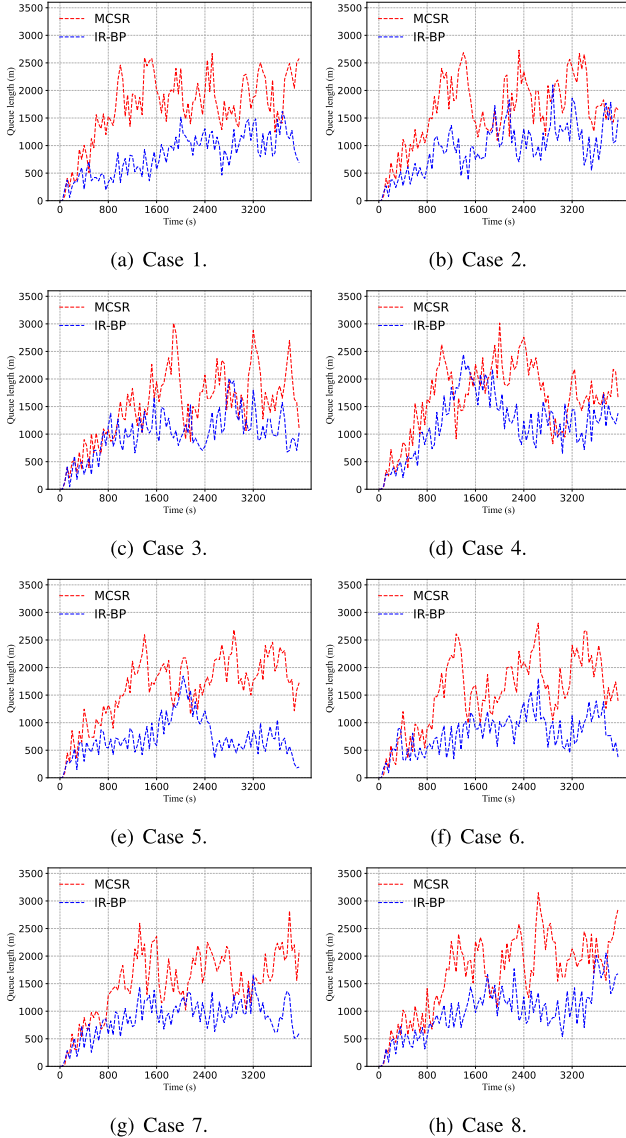


Fig. 14. Vehicle queuing length for different cases.

distances and travel times while reducing waiting time. This conclusion is further supported by Fig. 14. Although the queue length under the IR-BP control policy may be longer or unstable in some cases, it consistently performs better than the MCSR method. These experiments confirm the efficacy of the proposed method in mitigating congestion and improving transportation efficiency.

VI. CONCLUSION

We propose an integrated traffic control policy for heterogeneous container terminal networks to reduce congestion. A BP-based VTR control method is employed to determine both the activation order and duration of intersection phases, ensuring smoother traffic flow. To further mitigate HDV-related uncertainties, we incorporate a BP-based dynamic routing method that adjusts CAV paths at each intersection using travel time and a heuristic distance constraint. Experimental results demonstrate that the proposed control policy significantly reduces queue lengths and average travel times, effectively alleviating congestion while ensuring collision-free navigation at unsignalized intersections. Moreover, the relaxation parameter in this policy allows flexible avoidance of long routes while maintaining reasonable travel times. In future work, we plan to extend the proposed method to address transport tasks with time constraints, which could enhance its applicability to more complex traffic management scenarios.

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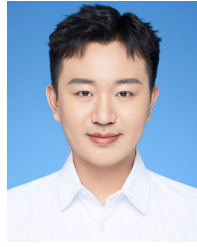
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